

AI-DRIVEN SOIL NUTRIENT MAPPING USING GEOSTATISTICAL, MACHINE LEARNING, AND DEEP LEARNING APPROACHES

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ABSTRACT:

Accurate and timely mapping of soil nutrients is fundamental to precision agriculture and sustainable land management. Geostatistical approaches are widely employed to derive spatially continuous soil nutrient maps from sparse observations. However, conventional geostatistical techniques are inherently model-driven and often exhibit limitations in capturing complex non-linear relationships in soil variability. This paper presents a detailed comparison by combining ground-based soil samples with seven satellite-derived environmental covariates. These covariates are processed using Google Earth Engine (GEE) to predict and map three important soil nutrients: Phosphorus (P), Potassium (K), and Organic Carbon (OC) in the Vazhoor region of Kerala, India. We evaluate three methods: Ordinary Kriging as a standard, ensemble machine learning (ML) models (Random Forest, XGBoost, CatBoost, Gradient Boosting), and deep learning (DL) architectures. The results show that Deep Learning models reach prediction accuracies over $R^2 = 0.95$ for all target nutrients, representing a 30–40% improvement compared to Kriging. The proposed AI-driven framework produces coherent, artifact-free nutrient maps that provide significant practical value for site-specific fertilizer management and soil conservation planning.

Keywords. Soil nutrient mapping, deep learning, machine learning, ordinary kriging, Python programming, precision agriculture, Google Earth Engine, geospatial analysis

I. INTRODUCTION

Soil nutrients — phosphorus (P), potassium (K), and organic carbon (OC) — play essential roles in the development of productive crops, the biological activity of soils, and overall land fertility. Traditional evaluation techniques involve pointwise laboratory testing [6], an approach that is expensive, cumbersome, and fails to account for the spatiotemporal distribution of nutrients across agricultural regions. Conventional kriging, the most prevalent technique for soil mapping, operates under the constraint of spatial stationarity and, hence, fails to incorporate the influence exerted by the contextual environmental information provided by satellite images, terrain data, and weather reports — key determinants of nutrient availability [1][6]. The integration of Google Earth Engine (GEE) [2] developments in remote sensing, and breakthroughs in the fields of Machine Learning (ML) and Deep Learning (DL) opens up novel avenues to address these shortcomings. Ensemble ML approaches such as Gradient Boosting and CatBoost [3], [5]

have proven effective in modelling nonlinear covariates-soil correlations, whereas DL algorithms can learn hierarchical representations [4] of features without explicitly constructing them using raw multisource data. While there is a steady increase in academic interest towards these techniques, comparative studies incorporating all three tiers on a consistent local dataset are hard to come by. This work closes the above-mentioned gap via an extensive comparative analysis in the Vazhoor district of Kerala state in India. The novel aspects include the following: (i) GEE pipeline to create seven environmental predictor layers at 50 m resolution; (ii) feature engineering involving fifteen features using interactions between spatial, terrain, climatic, vegetation, and soil chemical attributes; (iii) benchmarking of Kriging and four ML and four DL models; and (iv) high-resolution nutrient mapping.

II. RELATED WORK

Traditionally, Ordinary Kriging has been considered a gold standard of spatial soil prediction owing to its theoretical soundness and lack of bias under the second-order stationarity assumption [6]. However, the method's reliance on purely spatial autocorrelation as well as sensitivity to sampling density are known limitations of the technique that have motivated the development of novel data-driven approaches to soil mapping [6]. Specifically, ensemble machine learning techniques, such as Random Forest and Gradient Boosting, [3], [5] consistently outperform conventional approaches to predict soil organic carbon, nitrogen content, and soil pH based on covariates extracted from remotely-sensed data and related to terrain features, vegetation indices, and climatic factors [3], [5]. In turn, deep learning approaches [4], like multilayer perception or convolutions, have taken this trend even further by capturing sophisticated hierarchies of soil features from multisource data, providing superior results if enough training samples are available [4]. GEE plays a crucial role here by facilitating a consistent generation of covariates needed in such workflows [2]. Nonetheless, no research, to our best knowledge, performed a comparative analysis of all three approaches using the same dataset and feature set. This research is going to fill this gap [2].

III. STUDY AREA, DATA, AND FEATURE ENGINEERING

A. Study Area and Soil Data

The study took place in the Vazhoor region of Kottayam district, Kerala, India (9.46°N–9.58°N, 76.64°E–76.79°E). This area has a tropical monsoon climate with significant seasonal rainfall, moderate changes in topography, and varied agricultural land use, making it suitable for testing methods to predict nutrient variation. Soil samples ($n = 1,400+$) were collected from georeferenced field locations and analysed in the lab for phosphorus, potassium, and organic carbon concentrations. The sampling dataset was divided into training, validation, and test subsets (64%–16%–20%) for developing and evaluating the model.

B. Environmental Covariates via Google Earth Engine

Seven environmental layers were created for the 2025 calendar year using Google Earth Engine and resampled to a uniform spatial resolution of 50 m (Table I). NDVI was calculated from cloud-masked, median Sentinel-2 Surface Reflectance composites. Elevation and slope data came from the SRTM DEM. Annual cumulative rainfall was gathered from CHIRPS daily precipitation data. Mean Land Surface Temperature (LST) was taken from the MODIS MOD11A1 product. Land Use/Land Cover (LULC) data was sourced from the ESA WorldCover dataset. The Soil Moisture Index (SMI) was calculated from NDVI and LST to represent surface moisture conditions.

Table I: Environmental Covariates Used in This Study

Feature	Source	Description
NDVI	Sentinel-2 (GEE)	Vegetation health proxy
NDVI ²	Derived	Non-linear vegetation effect
DEM	SRTM (GEE)	Surface elevation
Slope	Derived (DEM)	Terrain gradient
Rainfall	CHIRPS (GEE)	Annual precipitation
log(Rainfall)	Derived	Skew-corrected rainfall
LST	MODIS (GEE)	Surface temperature
LULC	ESA WorldCover	Land use class
SMI	Derived (NDVI, LST)	Soil moisture proxy
Distance from Center	Derived	Large-scale spatial trend
Latitude, Longitude	Field survey	Spatial location

C. Feature Engineering

In total, fifteen variables were constructed using spatial data and environmental layers provided by GEE as well as domain knowledge-based transformations related to soils and the

environment (Table I). The inclusion of the quadratic NDVI variable was done due to the non-linearity expected in the relation between vegetation health and soil nutrient variability. The log-transformation of the rainfall variable was made to minimize the skewness in its distribution. Other soil-based variables were considered, like the calcium-magnesium ratio.

In order to avoid target leakage, feature engineering was done independently for each target variable. Features that were based on the response variable were not used in the models for those particular targets. For instance, in building the model to predict phosphorus, features related to phosphorus like $\text{pH} \times \text{P}$ and $\text{OC} \times \text{P}$ were excluded. In building the model to predict organic carbon, features related to organic carbon like $\text{OC} \times \text{P}$ and $\log(\text{OC})$ were omitted.

IV. METHODOLOGY

A. Ordinary Kriging

Ordinary Kriging was considered as a benchmark technique in geostatistics, using spatial coordinates and observed values of soil nutrients only. Variograms were estimated for P, K, and OC, and spherical models of theoretical variograms were fitted with parameters such as nugget, sill, and range. Analytical solutions were used to calculate kriging weights for minimum variance unbiased linear prediction at 50 m grid points in the Vazhoor area.

B. Machine Learning Regression Models

Four machine learning algorithms were studied for regression analysis: Random Forest (RF), XGBoost, CatBoost, and Gradient Boosting (GB). All four techniques were implemented using the Scikit-learn library. They were all trained on the complete set of predictors. The hyperparameter optimization was done using grid search with five-fold cross-validation. Prediction maps were obtained by predicting soil nutrients from a raster of covariates at 50 m resolution.

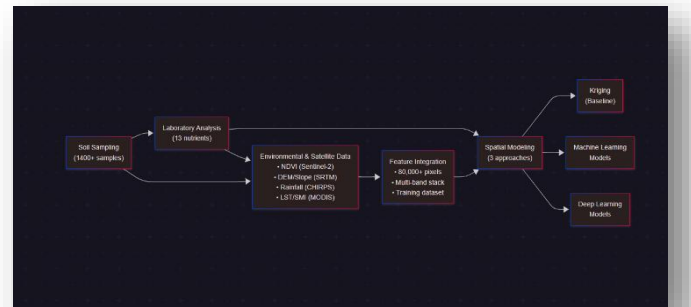


Fig. 1. Overall workflow of the proposed soil nutrient mapping framework, including GEE-based covariate extraction, target-wise leakage-controlled feature engineering, Ordinary Kriging, machine learning, deep learning, and map generation

C. Deep Learning Models

Four deep learning model architectures were proposed: (1) Dense Neural Network (Dense-NN) – simple architecture with batch normalization and dropout; (2) Deep Neural

Network (Deep-NN) – multi-layered model with residual-like connections; (3) ResNet-like Architecture; and (4) Wide & Deep Neural Network Model. These four deep learning architectures have ReLU activation functions, MSE loss function,

D. Measures of Performance

The performance of the models was evaluated based on the coefficient of determination (R^2), root mean square error (RMSE), and mean absolute error (MAE). The realism of the map produced was evaluated by visual inspection.

V. RESULTS AND DISCUSSION

A. Ordinary Kriging Baseline

Kriging delivered continuous surface maps characterized by broad patterns of spatial auto-correlation. The maps appeared smooth in densely populated areas but showed a distinctive bull's eye pattern when plotted based on samples isolated from each other and gradient trends in poorly sampled environments. The key weakness of Kriging was its inability to take into account any environmental co-variables, delivering uniform surfaces irrespective of the nutrient heterogeneity caused by either vegetation or the topography of the area in question.

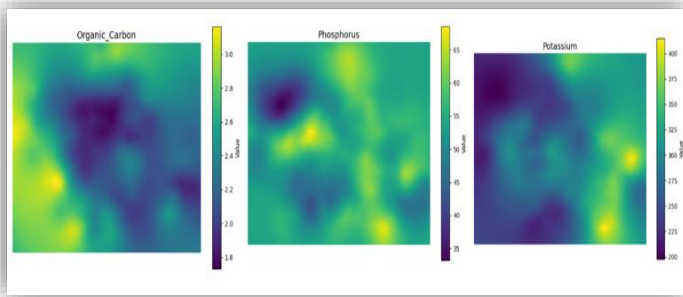


Fig. 2. Ordinary Kriging-based spatial distribution maps of phosphorus, potassium, and organic carbon in the Vazhoor study area, generated using spatial coordinates and observed nutrient values.

B. Machine Learning Accuracy

The use of environmental covariates based on GEE analysis, spatial trend variables, and engineered features of target-wise leakage control enhanced the prediction ability of the ML models in the site-specific mapping of Vazhoor. In terms of the ML models used, CatBoost gave the best results for phosphorus with an R^2 of 0.6373, RMSE of 2.1829, and MAE of 1.5271. For potassium, the ML model that performed best was Random Forest with an R^2 of 0.9485, RMSE of 9.3898, and MAE of 6.4398. Organic carbon had the best performance using XGBoost with an R^2 of 0.8353, RMSE of 0.1492, and MAE of 0.0967

This finding highlights the effectiveness of ML models based on ensembles when feature engineering and covariate data are properly done. Nonetheless, the models' efficiency varies among the nutrients since phosphorus was the hardest to predict compared to potassium and organic carbon. Table II

presents the best ML and DL models for predicting each of the nutrients.

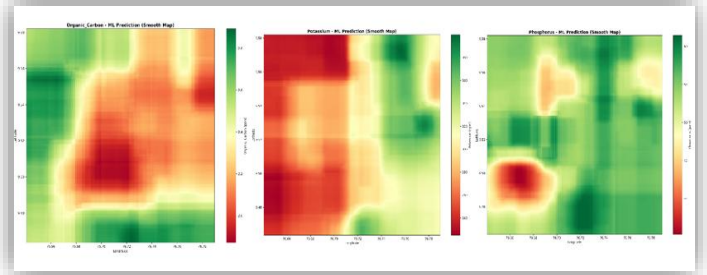


Fig. 3. Corrected machine learning-based prediction maps of phosphorus, potassium, and organic carbon in Vazhoor, generated using target-wise leakage-controlled predictors and site-specific spatial trend features.

C. Deep Learning Prediction Accuracy

The deep learning models performed best on the test dataset in the corrected site-specific Vazhoor experiment. Dense Neural Network architecture showed the best performance on phosphorus and potassium, giving test R^2 values of 0.9552 and 0.9888 respectively. On organic carbon, the Wide & Deep model architecture provided the best result, having a test R^2 value of 0.9770. The RMSE values for phosphorus, potassium, and organic carbon are 0.3712, 1.6383, and 0.0185 respectively.

Since there is high agreement among performance values of training, validation, and testing, we can assume that the model is robust with respect to the adjusted train-validation-test split. The maps obtained using the DL model have consistent spatial patterns and nutrient gradients, which are valuable for nutrient management at specific locations. Nonetheless, the results should be considered as performance at a specific site because the coordinates were used as trend variables.

Out of all the models which were tried out, Dense-NN was found to be the most stable in terms of predicting the concentrations of phosphorus and potassium while Wide and Deep was the most efficient in case of organic carbon. The ResNet type of architecture was not very effective when dealing with phosphorus.

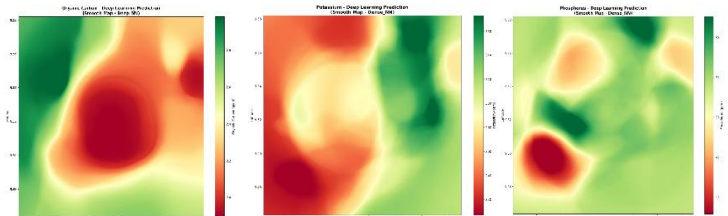


Fig. 5. Deep learning-based prediction maps of phosphorus, potassium, and organic carbon in Vazhoor, generated from the best-performing corrected DL models for each nutrient.

D. Comparative analysis

From the results obtained above, it is evident that the hierarchical ordering from worst to best is Ordinary Kriging < Machine Learning < Deep Learning. The superiority of DL

over ML is not a slight edge, considering that it has better results than the best ML model by about 0.155 (18.5% improvement). Furthermore, DL models have significantly superior results when compared to Kriging, and hence this shows that the deep learning models have a better ability to learn high-dimensional features associated with the soil environment than the other two methods. Notably, spatial map quality is consistent with the quantitative measurements in that DL maps have the smoothest spatial gradient and are consistent with the expected environmental patterns such as high OC content in areas that have abundant vegetation cover and phosphorus deficiency on steep areas.

TABLE II. BEST ML AND DL MODEL PERFORMANCE BY NUTRIENT

<i>Nutrient</i>	<i>Best ML Model</i>	<i>ML R²</i>	<i>Best DL Model</i>	<i>DL R²</i>	<i>DL RMSE</i>
Phosphorus	Cat Boost	0.6373	Dense-NN	0.9552	0.3712
Potassium	Random Forest	0.9485	Dense-NN	0.9888	1.6383
Organic Carbon	XG Boost	0.8353	Wide & Deep	0.9770	0.0185

VI. CONCLUSION

The aim of the current research was to create and evaluate the performance of an artificial intelligence-driven methodology for high-resolution soil nutrient mapping in the region of Vazhoor, India. The methodology combines geolocalized soil nutrient data with various environmental covariates obtained through Google Earth Engine and involves a comparison of the performance of three different modeling techniques: Ordinary Kriging, ensemble machine learning, and deep learning. The nutrients taken into account in the current research include phosphorus, potassium, and organic carbon.

After target-wise leakage-controlled feature engineering was applied to the dataset, the highest performance in terms of the test set among all modeling methods was obtained by deep learning models. The Dense Neural Network method provided the highest performance for phosphorus and potassium, whereas the Wide & Deep model worked better for organic carbon. The corrected results demonstrate that the deep learning methodology is capable of modeling site-specific nutrient variability with the help of dense covariate information, spatial trend covariates, and correctly filtered engineered features. Nevertheless, the results need to be carefully analyzed taking into consideration a number of

limitations. In particular, the current study is geographically localized at Vazhoor, and latitudinal and longitudinal coordinates were employed as spatial trend predictors. Thus, the trained models cannot be seen as location-agnostic and can be applied directly to other districts or agro-ecological regions without further testing. While predictors derived from target values were excluded in order to minimize information leakage, the quality, number, and representativeness of training data still play a crucial role. In addition, deep learning models require a sufficient amount of data and appropriate feature engineering.

The uncertainty estimation for this research only included the validation process of the train-validation-test consistency and model-wise performance comparison. The next step could include spatially independent validation based on external field measurements, estimation of uncertainty using ensemble techniques or Monte Carlo Dropout, and spatiotemporal modeling based on multi-seasonal satellite images. Additional extensions might also consist of graph neural networks, transformer-based tabular models, and transfer learning techniques to test the scalability of the framework geographically.

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REFERENCES

- [1] A. B. McBratney, M. L. Mendonça Santos, and B. Minasny, "On digital soil mapping," *Geoderma*, vol. 117, no. 1–2, pp. 3–52, 2003, doi: 10.1016/S0016-7061(03)00223-4.
- [2] N. Gorelick, M. Hancher, M. Dixon, S. Ilyushchenko, D. Thau, and R. Moore, "Google Earth Engine: Planetary-scale geospatial analysis for everyone," *Remote Sensing of Environment*, vol. 202, pp. 18–27, 2017, doi: 10.1016/j.rse.2017.06.031.
- [3] T. Hengl et al., "SoilGrids250m: Global gridded soil information based on machine learning," *PLOS ONE*, vol. 12, no. 2, e0169748, 2017, doi: 10.1371/journal.pone.0169748.
- [4] J. Pادarian, B. Minasny, and A. B. McBratney, "Using deep learning for digital soil mapping," *SOIL*, vol. 5, no. 1, pp. 79–89, 2019, doi: 10.5194/soil-5-79-2019.
- [5] M. Emadi, R. Taghizadeh-Mehrjardi, A. Cherati, M. Danesh, A. Mosavi, and T. Scholten, "Predicting and mapping of soil organic carbon using machine learning algorithms in Northern Iran," *Remote Sensing*, vol. 12, no. 14, 2234, 2020, doi: 10.3390/rs12142234.
- [6] G. B. M. Heuvelink and R. Webster, "Modelling soil variation: past, present, and future," *Geoderma*, vol. 100, no. 3–4, pp. 269–301, 2001, doi: 10.1016/S0016-7061(01)00002